SULIT

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER I , 2017 / 2018

SUBJECT CODE	:	**SCSJ 2013**
SUBJECT NAME	:	**DATA STRUCTURE AND ALGORITHM**
SECTION	:	**ALL**
TIME	:	**9.00 AM – 11.30 AM**
DATE/DAY	:	**11 / 01 / 2018 (THURSDAY)**
VENUES	:	**KTHO**

INSTRUCTIONS :

This test book consists of two parts (2) parts. Answer **ALL QUESTIONS.**

PART A	-	**10 Objective Questions**	**(15 marks)**
PART B	-	**5 Structured Questions**	**(85 marks)**
		Question 1 (Searching)	**(15 marks)**
		Question 2 (Linked List)	**(15 marks)**
		Question 3 (Stack)	**(20 marks)**
		Question 4 (Queue)	**(15 marks)**
		Question 5 (Tree)	**(20 marks)**

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **TEN (10)** printed pages excluding this page.

PART A – OBJECTIVE QUESTIONS **[15 marks]**

Part A consists of 10 Objective Questions. Choose the correct answer and write your answer in the test booklet. Each question carries 1.5 marks.

1. Which of the following is the **average case** in linear search algorithm?

   A. When the searched item is located in the middle of the array.
   B. When the searched item is not in the array at all.
   C. When the searched item is located at the last element in the array.
   D. When the searched item is located at the beginning of the array.

2. Binary Search algorithm can **ONLY** be applied to:

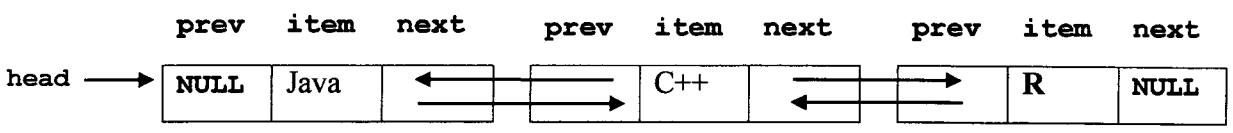
   A. Sorted linked list.
   B. Sorted array.
   C. Unsorted linked list.
   D. Unsorted array.

3. Which of the following statements are **TRUE** about linked list?

   i. Linked list implementation requires manipulation of pointer variable.
   ii. A series of items stored inside array variable can be considered as linked list.
   iii. Circular linked list is implemented if the next pointer for the last node point to the first node of the list.
   iv. Singly linked list is implemented if each intermediate node is pointing to other two nodes.

   A. i & ii
   B. ii & iv
   C. i & iii
   D. iii & iv

4. What is the type of linked list structure shown in the figure below?



   A. Singly linked list.
   B. Doubly linked list.
   C. Circular linked list.
   D. Circular doubly linked list.

5. After the following codes for stack execute, what will be the latest value for **top**, if **createStack()** function set **top** to -1 ?

```
Stack s;
s.push(4);
s.pop();
s.pop();
s.push(3);
s.pop();
```

A. -1
B. 0
C. 1
D. 2

6. Which of the following tasks **CANNOT** be solved using stack?

A. Manage the backtracking in search algorithm
B. Manage request for CPU
C. Manage the undo operation in the text editor tool
D. Manage the call of recursive function in a computer program

7. Which of the following is **NOT** a function in a queue circular linked list implementation?

A. **isEmpty()**

B. **isFull()**

C. **getFront()**

D. **getRear()**

8. Which of the following statement is **FALSE** about rightward drifting in queue?

A. It can be prevented using circular queue approach.

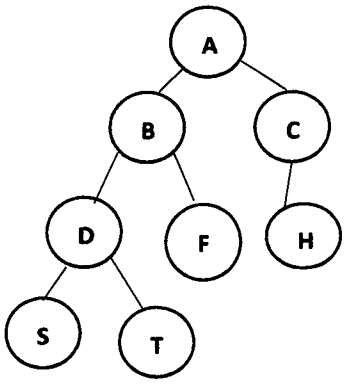
B. It may occur in queue using linked list.

C. It may happen after several deletion and insertion operations are done in the queue.

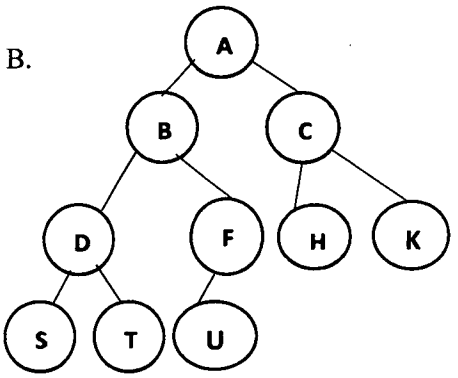
D. It can be avoided by shifting elements after every deletion in queue.

9. Which of the following binary tree is **Complete**?

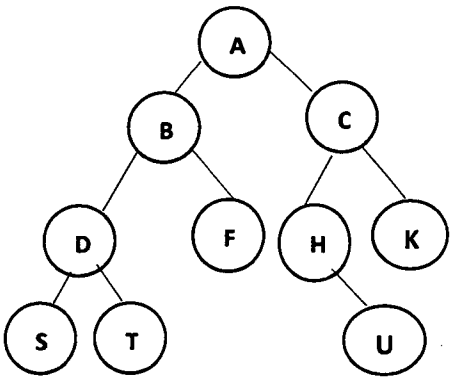
A.



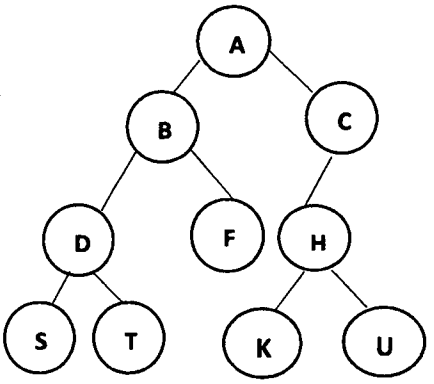
B.



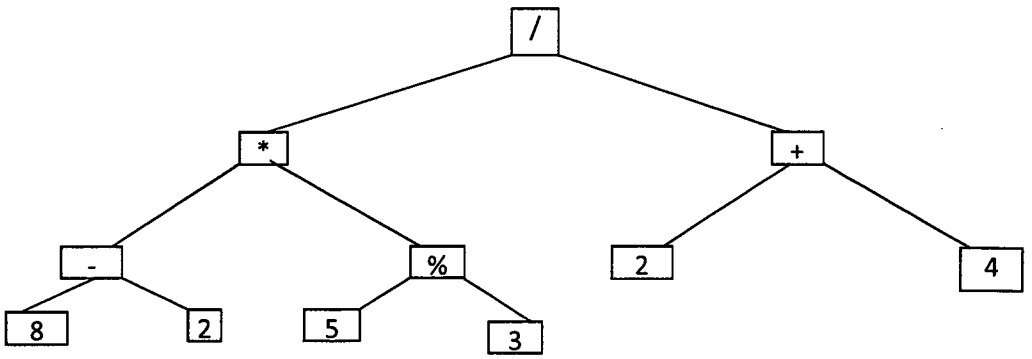
C.



D.



10. What is the **result of algebraic expression** represented by the expression tree in the figure?



A. 0
B. 2
C. 3
D. 1

PART B - STRUCTURED QUESTIONS **[85 marks]**

Part B consists of 5 structured questions. Answer all questions in the space provided. The marks for each part of the question is as indicated.

Question 1 **[15 marks]**

Given the linked list that contain information for books as in Figure 1a, answer question 1 (a).

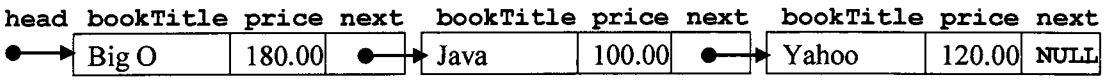


Figure 1a. Linked List of books

a) The linked list of books in Figure 1 is sorted in ascending order based on the book title. Name the searching technique that is suitable to be performed on the linked list if searching is done based on the book title. Give the reason to justify your answer.

[3 marks]

Given **NUM** array of integers as in Figure 1b, answer question 1(b) to 1(e).

index	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
	1	3	5	7	9	11	13	17	19

Figure 1b. NUM array

b) Perform Binary search for searching key **19** on the **NUM** array in Figure 1b. Show the tracing of your search using variables **left, right, middle, NUM [middle]** and **found** as shown in the table format below.

[3 marks]

left	right	middle	NUM[middle]	found

c) Perform Binary search for searching key **1** on the **NUM** array in Figure 1b. Show the tracing of your search using variables **left, right, middle, NUM[middle]** and **found** as shown in the table format below.

[3 marks]

left	right	middle	NUM[middle]	found

d) Fill in the number of comparisons and the complexity time when searching for key at the front array (1), middle array (9) and end of array (19) on **NUM** array in Figure 1b using **Binary search** and **Sequential search** technique.

[4 marks]

Searching Comparisons				
Search Key	Linear Search		Binary Search	
	Number of Compares	Complexity	Number of Compares	Complexity
1				
9				
19				

e) Discuss the efficiency analysis for the 3 cases based on the differences between the number of steps and the complexity time when searching for key at the front array (1), middle array (9) and at the end of array (19) using **binary search** and **sequential search** technique.

[2 marks]

Question 2 **[15 marks]**

`Program-2.1` is the implementation of `Position` class in C++. Answer questions 2(a) – 2(c).

```
1  // Program-2.1
2
3  class Position {
4      private:
5          float lat;  // lattitude
6          float lon;  // longitude
7          Position *next;
8          //.....
9      public:
10         Position(float pLat, float pLon) {
11             lat = pLat; lon = pLon; next = NULL;
12         }
13
14         void setNext(Position *newPost) { next = newPost; }
15         Position *getNext(void) { return next; }
16
17         void printPosition(void) {
18             cout << "(" << lat << ", " << lon << ")" << endl;
19         }
20 };
```

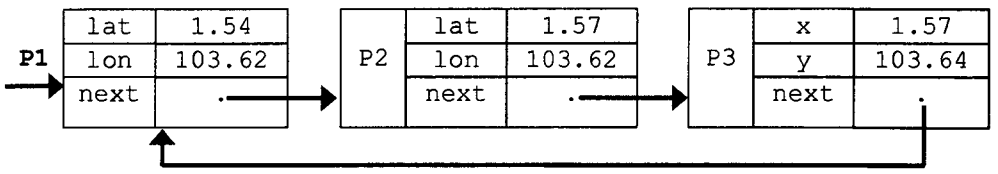


Figure 2a : Linked list of Position nodes

a) Write the instructions to form linked list structure of `Position` instances as in Figure 2a.

[6 marks]

```
Position *P1 = _____;  // creating new node pointed by P1

Position *P2 = _____;// creating new node pointed by P2

Position *P3 = _____;// creating new node pointed by P3

// Write codes to connect the 3 nodes based the structure in Figure 2a

_____;

_____;

_____;
```

b) Assume that the linked list structure of `Position` instances have been successfully constructed in Question 2 (a). Draw new linked list structure of `Position` instances if the following instructions are performed:

[3 marks]

```
Position *P4 = new Position(1.54, 103.64);
P4->setNext(P3->getNext());
P3->setNext(P4);
```

c) Add new lines of code after lines no. 7 to create a node for linked list structure as in Figure 2b. of **`Position`** instances can be constructed.

[1 mark]

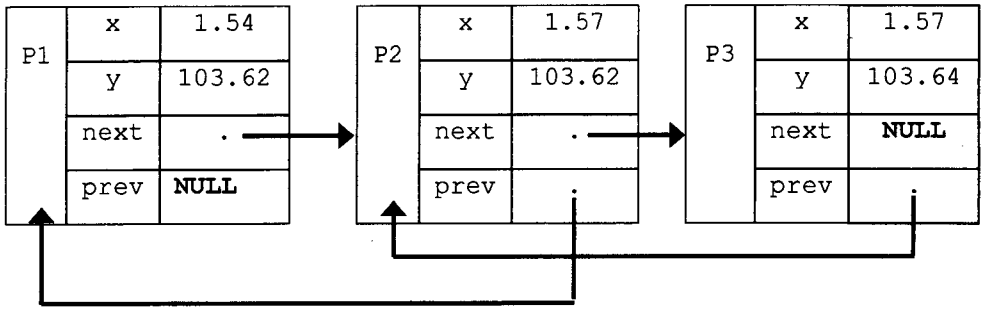


Figure 2b : Linked list of Position nodes

d) Write codes to connect the 3 nodes pointed by **`P1`**, **`P2`** and **`P3`** based the structure in Figure 2b.

[5 marks]

Question 3 **[20 marks]**

a) Give **TWO advantages** of stack implemented using linked list. [2 marks]

b) Convert the infix expression **(A + 5) * (B / C) – (2 * D)** into prefix and postfix expressions following the precedence, association and parentheses rules. Show the steps.

[6 marks]

Given Algorithm 3.1, answer Question 3©

```
//Algorithm 3.1 - check parentheses

create(stack);
continue=true;
while ((not end of input expression) && continue){
      ch=getch();
      if ((ch=='(') || (ch==')')){
            if (ch=='(')
                  push(stack, '(');
            else if (isEmpty(stack))
                  continue=false;
            else
                  pop(stack);
      }//endif
}//endwhile
if (end of input && isEmpty(stack))
      cout<<"the parentheses is balance";
else
      cout<< "the parentheses is not balance";

//end of algorithm
```

c) Based on the Algorithm 3.1 for checking the balance parentheses using stack, show that the expression **P – (Q / ((R + S - M))* T)** is balance. Show the checking process by completing the table below:

Input Ch	Content of Stack

[6 marks]

d) Show the steps of evaluating the postfix expression **5 4 * 3 2 + - 6 +**. Use the given format to write your answer. [6 marks]

Postfix	ch	operator	Operand1	Operand2	result	stack

Question 4 **[15 marks]**

a) Explain the differences of the following approach:
   i) Linear array based queue vs circular based queue.
   ii) Linear linked list based queue vs circular linked list based queue.

[5 marks]

b) Figure 4a shows the current content of a queue implemented using linear array.

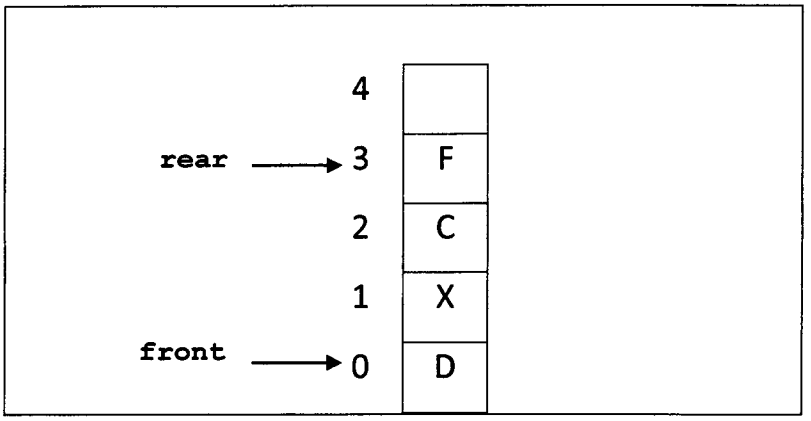


Figure 4a : Queue1 implemented using linear array

i) Redraw Queue1 (show the location of **`front`** and **`rear`**) in Figure 4a when character G is added (enQueue) into the queue.

[2 marks]

ii) Redraw Queue1 (show the location of **`front`** and **`rear`**) when deQueue is executed (queue state is a continuation from question b-i)

[2 marks]

iii) Redraw Queue1 (show the location of **`front`** and **`rear`**) when character P is added (enQueue) into the queue (queue state is a continuation from question b-ii)

[2 marks]

iv) Name and explain the phenomenon that is happening to Queue1 after deQueue and enQueue process in question (b-ii and b-iii). Why is this phenomenon a disadvantage to a queue?

[2 marks]

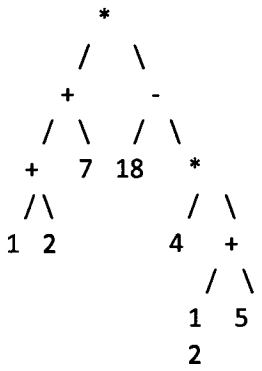
v) Suggest one approach to solve this problem and redraw Queue1 (show the location of **`front`** and **`rear`**) in Figure 4a when queue operations in b-i to b-iii are executed to solve the phenomenon based on your suggestion.

[2 marks]

Question 5 **[20 marks]**

a) Give the inorder, preorder and postorder traversal on the following expression tree.

[6 marks]



b) Draw a binary search tree if the following sequence of input values were inserted:

18, 22, 20, 9, 11, 15, 10, 6, 23, 5, 8 [4 marks]

c) Identify the type of the binary search tree drawn in question 5(b) (either complete, full, balanced). [2 marks]

d) Discuss the advantage of a balanced binary search tree and disadvantage of unbalanced binary search tree in terms of searching complexity.

[2 marks]

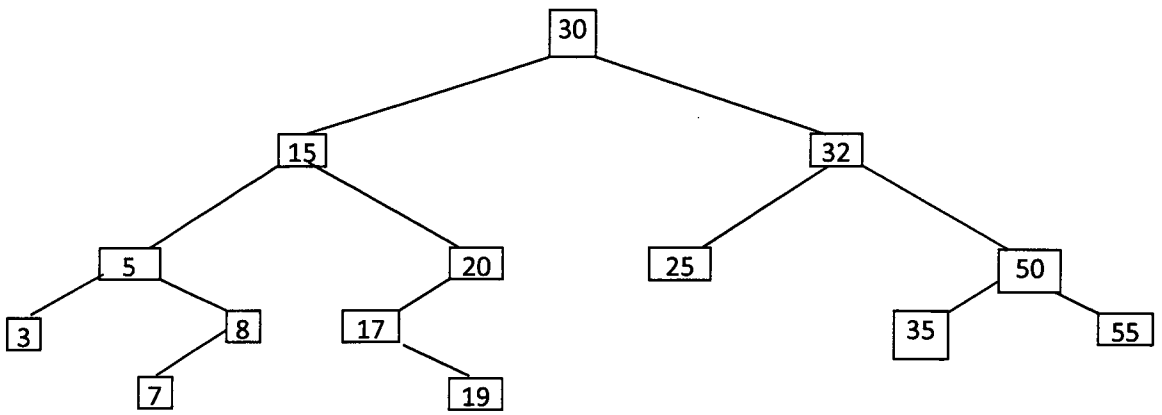


Figure 5a: Binary Search Tree of int value

e) Given the binary search tree in Figure 5a, redraw the tree after the following nodes are deleted **in sequence**. Show the new tree after every deletion.

i. Delete 17
ii. Delete 30
iii. Delete 15 [6 marks]